Density estimation and regression analysis on hyperspheres in the presence of measurement error

Junseok Lee

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1 The problem

Most previous work on hyperspherical data does not address measurement error. Some studies consider measurement error only on particular spheres. This paper develops deconvolution density estimation and regression analysis on a general unit hypersphere $\mathbb{S}^d$.

2 The setup

The latent variable X is not observed directly. Instead, we observe

$$Z = UX,$$

where U is an unobservable measurement error taking values in $\mathrm{SO}(d+1)$.

The first goal is to estimate the density f_X from i.i.d. observations $Z_1, \ldots, Z_n$. The second goal is to estimate the regression function

$$m : \mathbb{S}^d \to \mathbb{R}$$

under the model

$$Y = m(X) + \epsilon, \qquad E(\epsilon \mid X) = 0,$$

from i.i.d. observations

$$\{(Y_i, Z_i) : 1 \leq i \leq n\}.$$

The model also assumes

$$U \perp (X, \epsilon).$$

3 The main idea

The measurement error problem on $\mathbb{S}^d$ is treated as a deconvolution problem using harmonic analysis. Since the spherical harmonics

$$\{B_q^l : l \in \mathbb{N}_0,\ 1 \leq q \leq N(d,l)\}$$

form an orthonormal basis of $L^2(\mathbb{S}^d)$, the target density has the expansion

$$f_X = \sum_{l=0}^{\infty} \sum_{q=1}^{N(d,l)} \phi_q^l(f_X) B_q^l.$$

Under the measurement error model $Z = UX$, the densities satisfy the convolution relation

$$f_Z = f_U * f_X.$$

At each harmonic degree l, this gives

$$\boldsymbol{\phi}^l(f_Z) = \widetilde{\boldsymbol{\phi}}^l(f_U)\boldsymbol{\phi}^l(f_X).$$

When f_U is known and $\widetilde{\boldsymbol{\phi}}^l(f_U)$ is invertible,

$$\boldsymbol{\phi}^l(f_X) = \left\{\widetilde{\boldsymbol{\phi}}^l(f_U)\right\}^{-1}\boldsymbol{\phi}^l(f_Z).$$

The unknown coefficients of f_Z are replaced by empirical averages. The infinite harmonic expansion is truncated at degree T_n to control variability. The resulting deconvolution density estimator is

$$\widehat{f}_X(x) = \frac{1}{n}\sum_{i=1}^{n} \mathrm{Re}\left\{K_{T_n}(x, Z_i)\right\},$$

where K_{T_n} is the truncated deconvolution kernel.

The regression estimator is

$$\widehat{m}(x) = \widehat{f}_X(x)^{-1}\frac{1}{n}\sum_{i=1}^{n} \mathrm{Re}\left\{K_{T_n}(x, Z_i)\right\}Y_i.$$

A key property is

$$E\left\{K_{T_n}(x, Z) \mid X\right\} = K^*_{T_n}(x, X).$$

Thus, in conditional expectation, the deconvolution kernel constructed from the contaminated observation Z behaves like a kernel evaluated at the latent variable X.

4 The main results

The paper derives convergence rates and asymptotic normality under three smoothness conditions. For both density estimation and regression, it constructs asymptotic confidence intervals based on asymptotic normality and empirical likelihood. Simulation studies and a real data analysis are used to examine the finite-sample performance of the proposed estimators.

5 My takeaways

- Why is it sufficient to use only the real part of $K_{T_n}(x, Z_i)$, and how does the conjugate structure of the harmonic expansion ensure that no information is lost?
- How do the smoothness conditions on the measurement error distribution affect the choice of T_n and the resulting convergence rate?

References

[1] J. M. Jeon and I. Van Keilegom. Density estimation and regression analysis on hyperspheres in the presence of measurement error. *Scandinavian Journal of Statistics*, 51(2):513–556, 2024. `https://doi.org/10.1111/sjos.12684`.